

On the Development of a Portable Programmable Logic Controller (PLC) Trainer

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Abstract

In today's reality where sharing equipment between students is a health concern, university courses may be forced to go fully online with short notice, and students may have to quarantine for periods of time, there is a need for a portable Programmable Logic Controller (PLC) trainer that can be assigned to a student for the semester. The portable PLC trainer allows students to have a satisfactory laboratory experience across various modes of instructional delivery.

PLCs are used to automate industrial equipment and processes and are frequently used in laboratory activities in an automation course. At a minimum, a portable PLC trainer should be compact, have protection for the trainer components, provide user interface input and output devices, and the PLC programming software should be available for students to install on their own personal computers.

This paper presents the design and development of a portable PLC trainer including the requirements for the trainer, component selection rationale, and fabrication methods. Assessment of course learning outcomes through laboratory activities during the first semester of PLC trainer usage is presented. The PLC trainer developed has momentary pushbuttons, indicator lamps, potentiometers, a voltmeter, and sufficient digital and analog inputs and outputs for input/output devices on the trainer as well as external devices connected to the trainer.

Introduction

The adjustments that universities have had to make to instructional delivery modes since the start of the pandemic in 2020 have impacted laboratory activities significantly. When our university went completely online in the spring of 2020, an online PLC simulator, PLC Fiddle [1] was used as a substitute for hands on PLC trainers. While this was a decent substitute given the short notice of going completely online, it was clear that students were missing out on the experience of programming a real PLC.

A portable PLC trainer was needed so that students could take it home if the university were forced to go online or if the student had to quarantine or isolate. Each student would have a particular trainer assigned to them so that the danger of cross contamination is minimized.

Since students could take the trainer home with them, it means that they must be able to install the PLC programming software on their own personal computer.

The design of the portable PLC trainer described in this paper was begun near the end of the spring 2020 semester and was completed in late summer 2020. Funds were approved and fabrication of twenty trainers was completed in early fall 2020. Other similar portable PLC trainers are [2] and [3].

Course

ENGR 382 SCADA (Supervisory Control and Data Acquisition) Systems Design is an upper-division course taught to students in the Engineering Department as an elective or required course, depending upon the program. The following are the course learning outcomes:

1. Understand common Industrial Automation concepts, methods, and control algorithms.
2. Understand sensors and actuators used in Industrial Automation tasks.
3. Design Piping & Instrumentation Diagrams (P&IDs) for simple process systems.
4. Measure process variables in response to process parameters and analyze the resulting process behavior.
5. Understand Programmable Logic Controller (PLC) components, signal interface methods, and applications.
6. Design and write PLC control programs. Recognize other control program language formats defined in the IEC 61131-3 standard.
7. Design and program suitable Human Machine Interface systems.
8. Understand common industrial networking topologies, protocols, and hardware.

Course learning outcomes one, five, and six are related to PLC programming.

The course has the following laboratory activities:

- Lab 1: Automation Device Specifications
- Lab 2: CLICK Programming Software Introduction
- Lab 3: Basic Ladder Logic Programming
- Lab 4: Sequential Programming and Looping
- Lab 5: Car Lot Management System
- Lab 6: Stop Lights
- Lab 7: LabVIEW
- Lab 8: Analog I/O
- Lab 9: Analog Calculator
- Lab 10: Reaction Timer
- Lab 11: Process Trainer Closed-Loop Control
- Lab 12: Process Control System Design

Eight of twelve laboratory activities involve programming a PLC that is part of a PLC Trainer. Lab 1, Lab 7, Lab 11, and Lab 12 are not related to PLC programming. See the Appendix for examples of typical labs.

Methods

Requirements for the portable PLC trainer are developed based on the following need: a PLC trainer is needed for students to take home if they must be quarantined. Experience with the previous PLC trainer is also used to develop the requirements.

Course learning outcomes one, five, and six are assessed with Lab 2 to Lab 6 and Lab 8 to Lab 10. The lab scores are analyzed and presented using R. Success for an assessment item is defined as the average student performance being greater than 70%.

Previous PLC Trainer

The previous PLC trainer used in the course is shown in Figure 1. The PLC is the IEC FT1A SmartAXIS Touch that includes a human machine interface (HMI) with a color touch screen. The PLC has eight sinking digital inputs, four sourcing digital transistor outputs, two analog inputs, and two analog outputs.

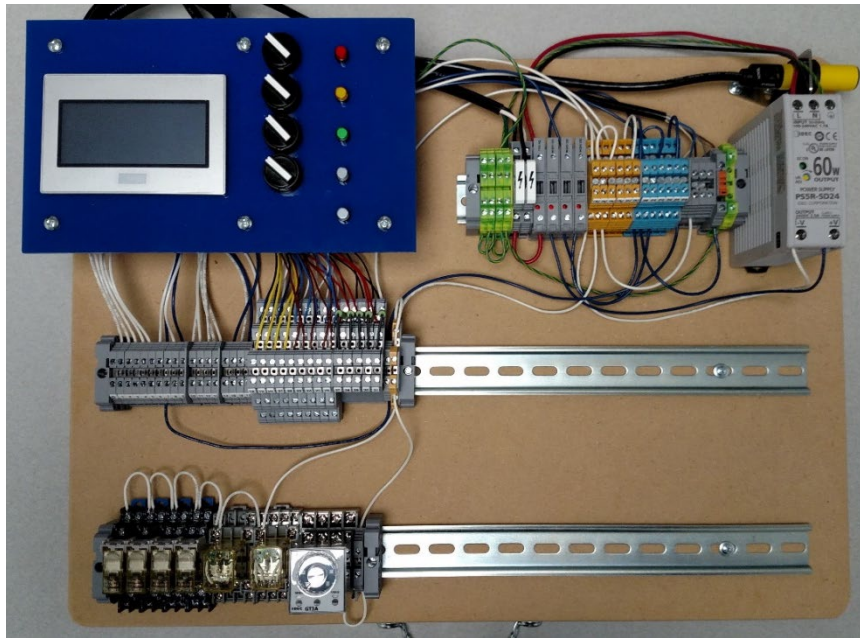


Figure 1: PLC Trainer with IEC PLC

The input devices include four two-position selector switches, five momentary push buttons, and an analog ultrasonic distance sensor. The output devices include five LEDs (inside the momentary push buttons), six relays, and a timer relay.

A 24 V DC, 60 W power supply powers all devices on the trainer.

All PLC inputs, PLC outputs, switches, push buttons, LEDs, the ultrasonic sensor, 24 V DC, and ground are connected to terminal blocks. All relays are mounted in sockets with screw terminals.

The trainer is large (16 inches [406 mm] by 22 inches [559 mm]) and heavy (14 pounds [6.4 kg]), so it is not very portable. Further, the programming software costs \$349 [4], so it is not reasonable to expect students to buy their own copy of the software.

Requirements for a Portable PLC Trainer

The requirements are based on the need expressed earlier, experience with the previous PLC trainer, and budget constraints. Due to the budget, a touchscreen HMI is not included in the requirements and will be added in the future. The HMI will connect to the PLC via Modbus TCP over an Ethernet connection.

The following are the requirements established for the portable PLC trainer.

1. The trainer must be portable (compact, not heavy, and have a handle).
2. The trainer components must be protected.
3. There must be user interface input and output devices (both digital and analog).
4. There must be sufficient digital and analog inputs and outputs for the internal user interface devices as well as an external module with sensors and actuators.
5. The PLC programming software must be available for students to install on their own personal computers.
6. The PLC programming software must be easy to use.
7. The PLC must support Modbus TCP communication. Modbus RTU communication is desired.
8. The cost should be minimized (less than \$1000 USD).

Portable PLC Trainer

Figure 2 shows the portable PLC trainer.

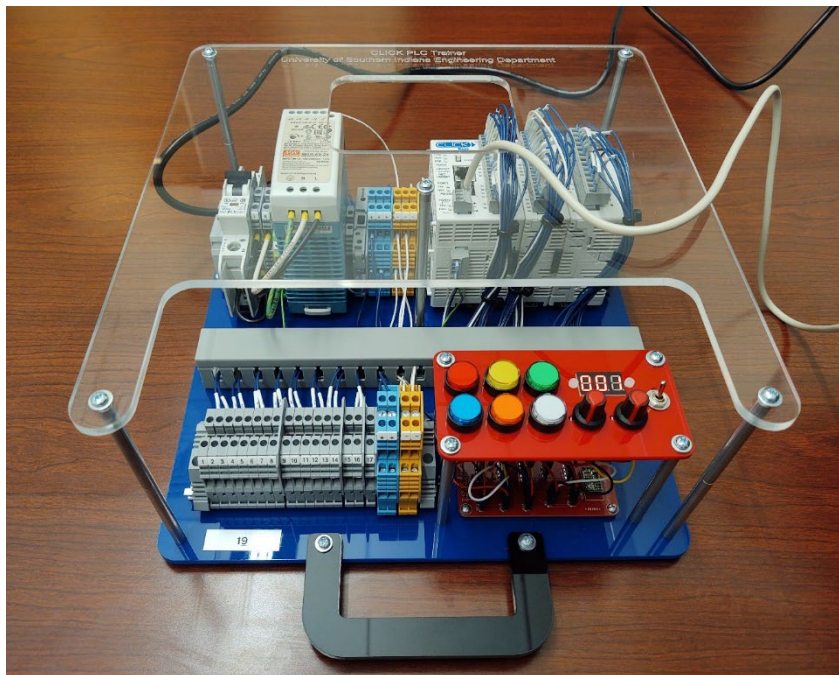


Figure 2: CLICK PLC Trainer

Table 1 defines the major components selected for the trainer.

Table 1: CLICK PLC Trainer Major Components

Component	Manufacturer	Model Number	Quantity
PLC	AutomationDirect	C0-12DD2E-2-D	1
Input Module	AutomationDirect	C0-16ND3	1
Relay Module	AutomationDirect	C0-08TR-3	1
Lighted Momentary Pushbuttons	IDEC	AL6M-M14-x	6
Potentiometers	-	10 k Ω	2
Voltmeter	-	0 V to 30 V DC	1
User Interface Board	Custom	-	1
Circuit Breaker	Eaton	FAZ-B1.5/1-NA-L	1
Power Supply	Mean Well	MDR-60-24	1

Most of the requirements affect the selection of the PLC. Requirement five implies that the PLC programming software must be free to install. This requirement alone eliminates quite a few PLC lines. Three PLC series from AutomationDirect were considered: CLICK [5], Productivity [6], and Do-more [7].

The CLICK series was able to satisfy all the requirements at the lowest cost.

Table 2 summarizes the input, output, and communication capabilities of the combination of the PLC, input module, and output module. In addition, the table lists the quantity of each capability type used directly on the trainer for user interface components (internal) and the quantity that is free for external use.

Table 2: PLC and Input/Output Module Capabilities

Capability Type	Quantity	Internal	External
High-Speed Digital Inputs (sinking)	4	0	4
Digital Inputs (sinking)	16	12	4
Digital Outputs (sourcing)	4	0	4
Relay Outputs	8	6	2
Analog Inputs (0 V to 10 V)	4	2	2
Analog Outputs (0 V to 10 V)	2	1	1
RS-232 Serial Communication	1	0	1
RS-485 Serial Communication (Modbus RTU)	1	0	1
Ethernet (Modbus TCP, Ethernet /IP)	1	0	1

The inputs and outputs not used internally are connected to terminal blocks so that external devices may be connected to the trainer (requirement four is satisfied).

The user interface (UI) has six lighted momentary pushbuttons (red, yellow, green, blue, orange (amber), and white) with normally open (NO) and normally closed (NC) contacts. Using lighted pushbuttons saves space over having separate lights and pushbuttons.

The analog portion of the UI has a toggle switch on the right-hand side of the UI that controls power to the analog devices. A switch-mode regulator steps down the 24 V to 10 V to power the potentiometers. One analog output on the PLC is connected to a 3-digit voltmeter that can measure 0 V to 30 V.

The devices in the user interface satisfy requirement 3.

Figure 3 shows a custom designed printed circuit assembly containing the switch-mode regulator and it provides connections between the user interface devices, power, inputs, and outputs.

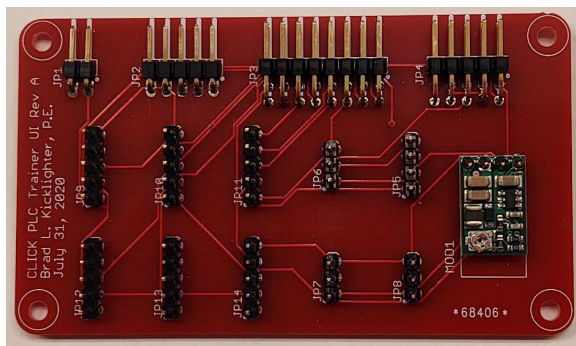


Figure 3: User Interface Printed Circuit Assembly

The trainer plugs into 120 V AC using the attached line cord. The trainer may be turned on by turning on the 1.5 A circuit breaker in the upper left corner of the trainer. This applies power to the 60 W, 24 V DC power supply that powers everything else on the trainer. Given the devices on the trainer and the anticipated external devices, the 60 W power supply is more than sufficient for the task.

There is a fuse holder to the right of the power supply that contains a 3 A fuse. Between the circuit breaker and the fuse, the overcurrent protection needs of the trainer are met.

The line cord is secured to the trainer with a cable tie (satisfies requirement two).

The base plate, top plate, handle, and user interface plate are acrylic that was cut out on a laser engraver. Threaded aluminum standoffs are used to attach the plates to each other. All components, except for the user interface, are mounted to 35 mm DIN rails. A wiring duct is used to route most of the wiring on the trainer.

The construction provides reasonable protection for the components (satisfies requirement two) while providing access where needed. The clear top plate makes all components visible to the user.

Not including the handle, the footprint of the trainer is less than 12 inches (305 mm) square. The trainer weighs less than six pounds (2.7 kg). Requirement one is satisfied.

Results and Discussion

In the fall of 2020, these PLC trainers were used for the first time with a class of sixteen students. Laboratory activities were used to assess course learning outcomes. Table 3 maps labs to outcomes.

Table 3: Mapping of Laboratory Activities to Course Learning Outcomes

Lab	Outcome 1	Outcome 5	Outcome 6
2		X	X
3		X	X
4	X	X	X
5	X	X	X
6	X	X	X
8	X	X	X
9	X	X	X
10	X	X	X

Lab scores from fall 2020 ($n = 16$ for each lab) are presented as box plots in Figure 4 and the average scores are tabulated in Table 4.

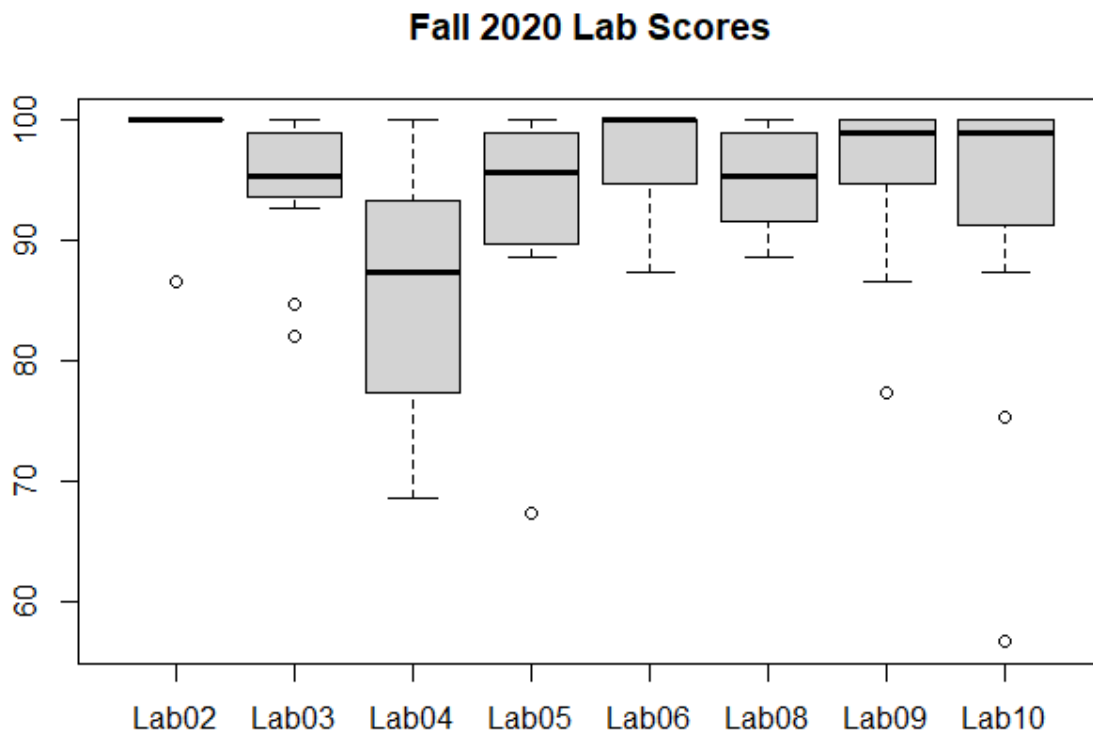


Figure 4: Box plots of lab scores from fall 2020

Table 4: Average lab scores from fall 2020

Lab	2	3	4	5	6	8	9	10
Average	99.2%	94.9%	85.1%	93.3%	97.5%	95.2%	96.1%	93.1%

The lab with the greatest variation is Lab 4. Since this lab introduces the concepts of sequential programming and looping, this is not surprising. There is a step change in difficulty from Lab 3 to Lab 4. Since the average on every lab is greater than 70%, the success criterion is satisfied for each lab.

Conclusions and Recommendations

The cost of trainers is less than \$600, so requirement eight is satisfied. The students found the trainers easy to use with very few issues. Figure 5 shows the trainer kit, which includes a small bag, flat blade screwdriver, and programming cable.

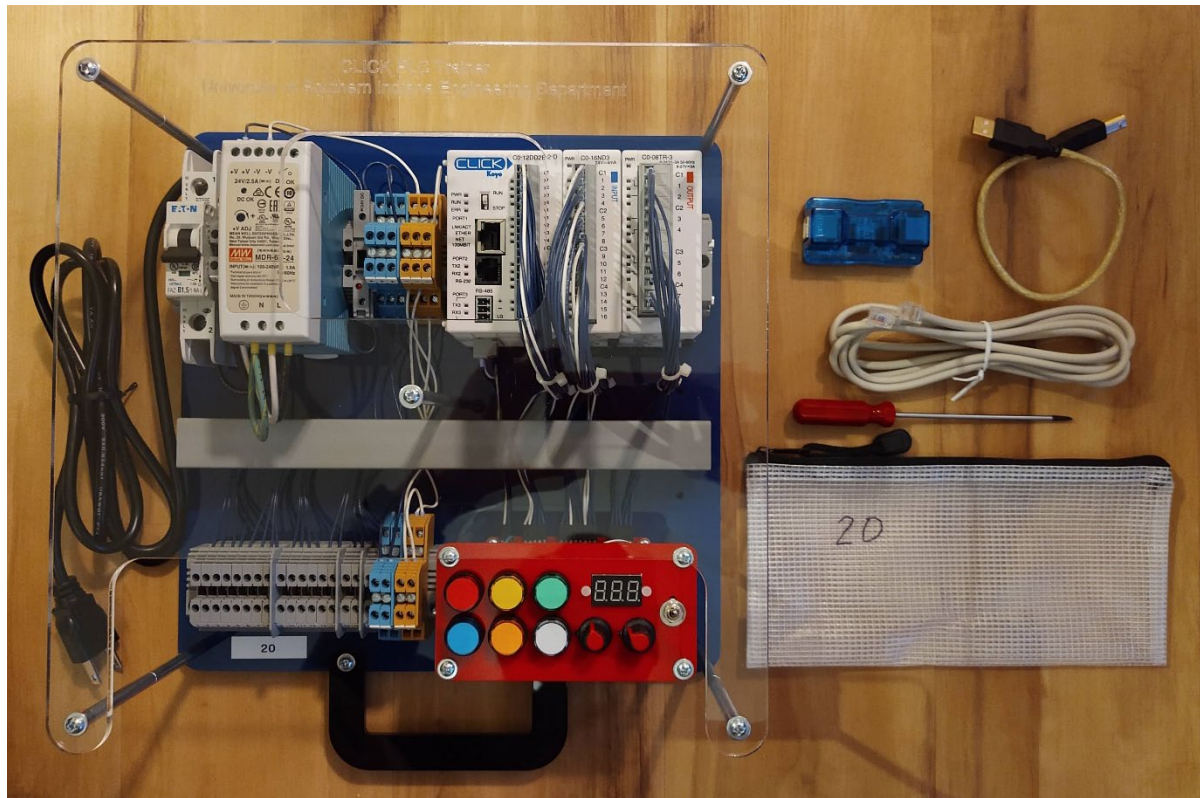


Figure 5: CLICK PLC Trainer Kit

Throughout the fall semester, there were instances where students took the trainers home to complete labs or because they had to quarantine. These events alone justified all the effort put into designing and fabricating the trainers.

All the requirements established for the trainer were met by this design.

All labs involving the trainer met their success criterion.

One thing that this trainer lacks that the previous one had is a touchscreen HMI. As mentioned previously, the HMI was not included due to budget reasons. It is planned to add an HMI as a standalone module that connects to the trainer via Ethernet.

Acknowledgements

I would like to thank my family and others for the help they provided in the assembly of the PLC trainers: Judy Kicklighter, Ben Kicklighter, Meg Kicklighter, Joan Kicklighter, Joyce Godeke, Harold Godeke, and Brayden McKinney.

References

- [1] "PLC Fiddle," *plcfiddle.com*. [Online]. Available: <https://www.plcfiddle.com/>. [Accessed: 22-Mar-2020].
- [2] A. Akparibo, A. Appiah, O. Fosu-Antwi, "Development of a Programmable Logic Controller Training Platform for the Industrial Control of Processes," *American Scientific Research Journal for Engineering, Technology, and Sciences (ASRJETS)*, vol. 15.1, pp. 186-196, Jan. 2016.
- [3] M. Barrett, "The design of a portable programmable logic controller (PLC) training system for use outside of the automation laboratory," *International Symposium for Engineering Education 2008, Dublin City University, Ireland*.
- [4] "Automation Organizer Software," *IDEC Corporation*. [Online]. Available: <https://us.idec.com/idec-us/en/USD/Software/Automation-Organizer/p/SW1A-W1C>. [Accessed: 28-Feb-2021].
- [5] "CLICK Series Programmable Controllers," *AutomationDirect*. [Online]. Available: https://www.automationdirect.com/adx/overview/catalog/programmable_controllers/click_series_plcs. [Accessed: 28-Feb-2021].
- [6] "Productivity Series Programmable Logic Controllers (PLCs)," *AutomationDirect*. [Online]. Available: https://www.automationdirect.com/adx/overview/catalog/programmable_controllers/productivity_series_controllers. [Accessed: 28-Feb-2021].
- [7] "Do-more Series Programmable Controllers," *AutomationDirect*. [Online]. Available: [https://www.automationdirect.com/adx/overview/catalog/programmable_controllers/do-more_series_\(brx,_h2,_t1h\)_plcs_\(micro_modular_-a-_stackable\)](https://www.automationdirect.com/adx/overview/catalog/programmable_controllers/do-more_series_(brx,_h2,_t1h)_plcs_(micro_modular_-a-_stackable)). [Accessed: 28-Feb-2021].

Appendix

Labs three and four are provided as examples of typical labs for this course.

Lab 3: Basic Ladder Logic Programming

Name:

Date:

Objectives

1. Demonstrate the ability to create simple ladder logic programs.
2. Demonstrate understanding of how logic gates are implemented in ladder logic.
3. Translate a Boolean expression into rungs in a ladder logic program.
4. Demonstrate understanding of the seal in and the seal breaker.

References

Ch. 21 Introduction to Programmable Controllers

Ch. 22 Fundamental PLC Programming

CLICK Programming Software – Getting Started

AutomationDirect C0-USER-M User Manual

AutomationDirect CLICK Programming Software Help File

Equipment

CLICK PLC Trainer

Programming Cable

PC with CLICK Programming Software

Thumb Drive or Cloud Storage

Procedure

For each part below, create a new project. Name your project files as “Lab3-<part>.ckp”. For Part 1, the project name will be “Lab3-1.ckp”.

Create a PDF (print to PDF) of each ladder logic program named “Lab3-<part>.pdf”. For Part 1, the PDF name will be “Lab3-1.pdf”.

For Parts 1, 2, and the Bonus, use the nicknames defined in Table 1. The nicknames for Part 3 are defined in that section.

Address	Nickname
X101	A
X103	B
X105	C
Y201	Out1
Y202	Out2
Y203	Out3
Y204	Out4
Y205	Out5

Table 1: Nicknames

For the truth tables, use the values 0 and 1, where 0 indicates the output is off and 1 indicates that it is on.

Part 1 – Logic Gates

As we learned in the lecture, digital logic gates can be implemented in ladder logic. You will create a ladder logic program with rungs that implement NOT, OR, NOR, AND, and NAND and then fill in the truth table for each of logic gates.

Create the ladder logic program shown in Figure 1. Use the nicknames defined previously and create rung comments as shown in Figure 1.

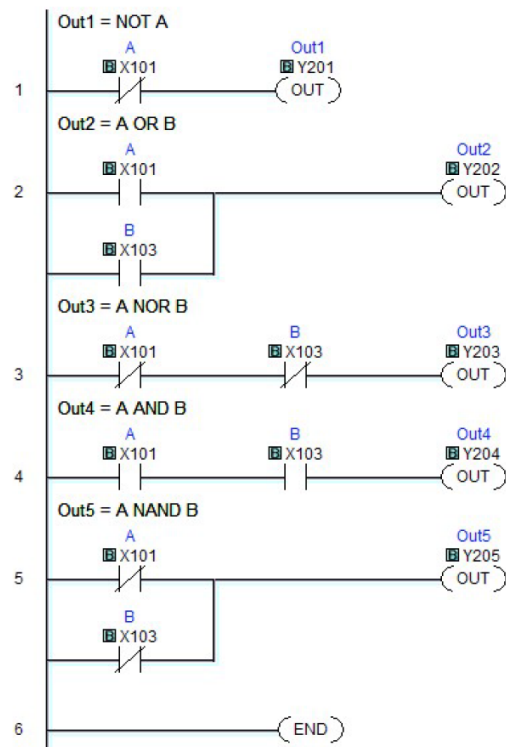


Figure 1: Logic Gates in Ladder Logic

Write your project to the PLC, test all input conditions, and fill in the truth tables (Tables 2 to 6).

A	Out1
0	
1	

Table 2: NOT Truth Table

A	B	Out2
0	0	
0	1	
1	0	
1	1	

Table 3: OR Truth Table

A	B	Out3
0	0	
0	1	
1	0	
1	1	

Table 4: NOR Truth Table

A	B	Out4
0	0	
0	1	
1	0	
1	1	

Table 5: AND Truth Table

A	B	Out5
0	0	
0	1	
1	0	
1	1	

Table 6: NAND Truth Table

Part 2 – OR/AND Combinations

Create a ladder logic program with two rungs. The first rung must implement the Boolean expression:

$$\text{Out1} = A \text{ OR } (B \text{ AND } C)$$

The parentheses indicate the precedence of the operations. In the first rung, B and C must be ANDed together and then the result is ORed with A.

The second rung must implement the Boolean expression:

$$\text{Out2} = A \text{ AND } (B \text{ OR } C)$$

Use the Boolean expressions as the rung comments in the program.

Write your project to the PLC, test all input conditions, and fill in the truth tables (Tables 7 to 8).

A	B	C	Out1
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Table 7: Rung 1 Truth Table

A	B	C	Out2
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Table 8: Rung 2 Truth Table

Part 3 – Start/Stop Logic

For this program, use the nicknames in Table 9.

Address	Nickname
X105	Start
X102	Stop
Y206	Motor

Table 9: Nicknames

Identify the input/output devices being used in the program. For the pushbuttons, identify the color and if it is normally open (NO) or normally closed (NC). For the LED, identify the color. Complete Table 10 with this information.

Nickname	Input/Output Device
Start	
Stop	
Motor	

Table 10: Input/Output Devices

Create the ladder logic program shown in Figure 2. Use the nicknames defined in Table 9 and create the rung comment as shown in Figure 2. Verify program operation to start and stop the motor.

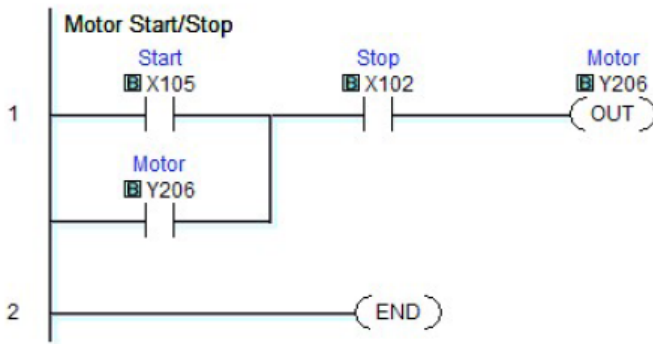


Figure 2: Motor Start/Stop Program

Identify the following program elements: seal in and seal breaker. Complete Table 11 with the nicknames.

Program Element	Nickname
Seal In	
Seal Breaker	

Table 11: Program Elements

Deliverables

Submit the following to Blackboard under the appropriate lab assignment:

- Edited lab document.
- Project and PDF files for all parts of the lab.

Lab 4: Sequential Programming and Looping

Name:

Date:

Objectives

1. Create a ladder logic program with initialization of a variable.
2. Create a ladder logic program using an on-delay timer.
3. Create a sequential ladder logic program to automate a set of steps.
4. Create sequential ladder logic program that loops.

References

Sequential Programming and Looping

CLICK Programming Software – Getting Started

AutomationDirect C0-USER-M User Manual

AutomationDirect CLICK Programming Software Help File

Equipment

CLICK PLC Trainer

Programming Cable

PC with CLICK Programming Software

Thumb Drive or Cloud Storage

Procedure

For each part below, create a new project. Name your project files as “Lab4-<part>.ckp”. For Part 1, the project name will be “Lab4-1.ckp”.

Create a PDF (Microsoft Print to PDF) of each ladder logic program named “Lab4-<part>.pdf”. For Part 1, the PDF name will be “Lab4-1.pdf”. Use the “letter” size and “landscape” orientation.

Each rung of the program must have a descriptive comment to document its behavior.

Demonstrate each of your programs to your instructor.

Answer the questions for each part in this document.

Part 1 – Sequential Program

Create nicknames for the following:

Address	Tag
X105	Start
X102	Stop
Y201	Step1
Y202	Step2
Y203	Step3
Y204	Step4
Y205	Step5
Y206	Step6
T1	Step1Done
T2	Step2Done
T3	Step3Done
T4	Step4Done
T5	Step5Done
T6	Step6Done
C1	Done
DS1	Timeout

Table 1: Nicknames

Create a sequential program with six steps and a done state.

Each step must last for 1.5 seconds.

Setpoints for all timers must be set to the Timeout data register.

Initialize Timeout upon first scan of the ladder logic program (see system coil relay SC2).

Use the On-Delay timer to provide the timing of each step (one timer for each step).

The completion of the timer for a step must cause the next step to start. The completion of the timer represents an event that signals that a particular step has completed.

Only one step may run at a time.

Go to the Done state after Step 6 is done.

The Start button must start Step 1 going.

The Stop button must stop any and all steps from going (including the Done state).

Download your program to the PLC and monitor it.

Demonstrate your program to the instructor.

Q1: What time base did you use for your timers?

A1:

Q2: What value did you use for Timeout?

A2:

Q3: Why would you want the ability to stop a PLC program?

A3:

Part 2 – Looping

Make a copy of your Part 1 project file and rename it for Part 2.

Modify the program so it will loop. That is, when Step 6 completes, Step 1 must start again.

The Done state is not needed.

Only one step may run at a time.

The Start button must start Step 1 going.

The Stop button must stop any and all steps from going.

Download your program to the PLC and monitor it.

Demonstrate your program to the instructor.

Q4: What did you have to do to the program for Part 2 to make it loop?

A4:

Q5: Why would you want a ladder logic program to loop?

A5:

Deliverables

Submit the following to Blackboard under the appropriate lab assignment:

- A text or Word document with the title “Lab 4”, your name, and the answers to the questions (A1 to A5).
- Project and PDF files for all parts of the lab.